

Sarah Connor ■

AI Security Engineer · Adversarial ML & Red Teaming

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> SUMMARY

AI security engineer with six years spanning applied machine learning, security engineering, and detection engineering. Builds defenses against adversarial and prompt-injection attacks on production ML systems, and leads red-team exercises against internal LLM deployments.

> EXPERIENCE

Senior AI Security Engineer

2023 – Present

Cyberdyne Systems

Fremont, CA

Leads detection engineering for production ML systems serving over ten million inference requests a day.

- Built an adversarial-example detector that cut false negatives by 40% against a red-team benchmark suite.
- Led quarterly red-team exercises against internal LLM deployments, uncovering prompt-injection paths later closed with input sanitization.
- Introduced model-serving anomaly scoring, reducing mean incident detection time from six hours to twenty minutes.

PyTorch · vLLM · LangChain · Ray

Security Engineer

2021 – 2023

Skynet Analytics

Remote

Owned security for the model-serving platform supporting the company's recommendation and fraud-detection products.

- Built anomaly-detection pipelines for model-serving infrastructure, catching two supply-chain compromises in third-party model artifacts.
- Automated red-team tooling for LLM jailbreak testing, cutting manual review time by 70%.
- Partnered with platform engineering to add authenticated model-artifact signing across the deployment pipeline.

Terraform · AWS · Python

Machine Learning Engineer

2019 – 2021

Tyrell Applied ML

Ljubljana, Slovenia

Built and shipped machine learning models powering fraud detection and demand forecasting for a fast-growing fintech product.

- Built and shipped fraud-detection models into production, reducing false-positive rates by 25% across the company's core payments pipeline.
- Designed the team's first automated retraining pipeline, cutting model refresh time from weeks to days.
- Mentored two junior engineers on feature engineering and model evaluation practices.

TensorFlow · scikit-learn · Airflow · Docker

> SKILLS

Security Burp Suite, Nmap, Metasploit, OSCP methodology, threat modeling

ML/AI PyTorch, vLLM, adversarial robustness, prompt-injection defense

Infra Kubernetes, Terraform, AWS, CI/CD pipeline hardening

Practice Incident response, secure SDLC, red-team exercise design

> EDUCATION

M.S. Computer Security

Stanford University

2021

Stanford, CA

B.S. Computer Science

University of Ljubljana

2019

Ljubljana, Slovenia (EU)

> PROJECTS

promptfirewall

github.com/sconnor/promptfirewall

2023

> CERTIFICATIONS, AWARDS & PUBLICATIONS

AI Village CTF – 1st Place

DEF CON

2024

Detecting Prompt Injection at Scale

arXiv preprint

2024

arxiv.org/abs/2403.09217

OSCP – Offensive Security Certified Professional

Offensive Security

2019